

AKHIL N MUTHYALA

Software Development Engineer-1

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PROFFESIONAL SUMMARY

Full Stack .NET Developer with hands-on experience designing scalable web applications using .NET, C#, ASP.NET Core, React.js, JavaScript, SQL Server, REST APIs, HTML, CSS, Bootstrap, and Azure DevOps. Skilled in React project development, reusable UI components, API integration, SQL optimization, debugging, CI/CD support, Agile delivery, AI-assisted coding, and legacy system modernization, including mainframe screens.

TECHNICAL SKILLS

Frontend: React.js, TypeScript, JavaScript, HTML5, CSS3, Bootstrap, jQuery, AJAX, Angular

Backend: C#, .NET, ASP.NET Core, ASP.NET MVC, Razor Pages, Web API, REST APIs, Node.js

Database: SQL Server, SQLite, SQL, Stored Procedures, Query Optimization

DevOps & Tools: Azure DevOps, Git, GitHub, CI/CD, Jira, Visual Studio, VS Code

Libraries & Frameworks: React Query, MobX, Axios, MediatR, Entity Framework Core, JWT

Accessibility & AI Tools: JAWS, NVDA, AI-Assisted Code Review, Claude Code, Prompt-Driven Development

EXPERIENCE

Software Development Engineer-1, Deloitte, Lexington-Fayette

Oct 2024 to Present

Project: Missouri Child Support System Modernization

Technologies: ASP.NET Core, Razor Pages, C#, JavaScript, HTML, CSS, Bootstrap, SQL Server, REST APIs, Azure DevOps, Git, NVDA, JAWS, AI-Assisted Development Tools

- Developed enterprise web applications using ASP.NET Core, C#, Razor Pages, JavaScript, SQL Server, HTML, CSS, Bootstrap, REST APIs, and Azure DevOps.
- Modernized 20+ legacy mainframe screens into secure, responsive web modules for the Missouri Child Support System, improving usability, accessibility, performance, and maintainability.
- Designed secure Razor Page Models, validation logic, reusable UI patterns, form controls, and data-driven features for case management workflows.
- Built dynamic UI features, including date pickers, masked inputs, time pickers, validation controls, modal overlays, and responsive layouts.
- Integrated web modules with legacy backend systems using proxy DLLs and REST APIs to support secure data exchange with SQL Server business logic.
- Optimized SQL Server queries, stored procedures, joins, and data-access logic to reduce latency and improve application performance.
- Implemented Section 508/WCAG accessibility improvements using NVDA and JAWS, including keyboard navigation, ARIA support, focus handling, and screen-reader fixes.
- Resolved UI, backend, API, and database defects while supporting Azure DevOps CI/CD pipelines, pull requests, deployments, and release tracking across Dev, QA/QC, and UAT.
- Collaborated with QA, business analysts, developers, and stakeholders to validate requirements, troubleshoot issues, maintain documentation, and deliver stable releases.
- Used AI-assisted development tools for debugging, documentation, code review support, unit test planning, and legacy code analysis.

Junior Developer, SITEK Inc, Lexington

Aug 2024 to Oct 2024

Project: Cumberland Surety

Technologies: React.js, TypeScript, ASP.NET Core Web API, C#, SQL Server, REST APIs, Entity Framework Core, Bootstrap, Azure DevOps, Git

- Developed a full-stack bond management web application using React.js, TypeScript, ASP.NET Core Web API, C#, SQL Server, and REST APIs.
- Built React modules for Bond Search, Bond Details, Agent Management, Reports, Payments, Inspection, Renewal/Billing, Collateral, Capital, and Data Tables.

- Created reusable React components, forms, dropdowns, search filters, validation logic, editable screens, action buttons, and data grids for surety bond workflows.
- Integrated React screens with ASP.NET Core Web API services and SQL Server data sources for record search, updates, reporting, and business logic.
- Supported backend development using C#, ASP.NET Core, Entity Framework Core, REST APIs, and SQL Server to manage bond, permit, agent, renewal, and payment data.
- Optimized SQL Server queries, stored procedures, joins, indexes, and data-access logic to improve performance and data retrieval speed.
- Collaborated with developers, QA, and stakeholders on debugging, testing, defect resolution, code reviews, Agile delivery, and Azure DevOps release support.

Graduate Assistant Programmer, Eastern Kentucky University, Richmond

Aug 2023 to May 2024

- Mentored Computer Science students in HTML, CSS, Bootstrap, JavaScript, TypeScript, Angular, C#, and object-oriented programming fundamentals.
- Led lab sessions and office hours to troubleshoot coding issues, explain course concepts, and guide students through programming assignments.
- Prepared lesson materials, technical demos, sample code, and exercises focused on responsive UI design, reusable components, validation, accessibility, and clean coding.
- Reviewed student code and provided feedback on debugging, Git workflows, Angular components, TypeScript, JavaScript logic, and C# concepts.
- Supported local development setup using Node.js, npm, Angular CLI, Git, Visual Studio, VS Code, and browser developer tools.

Software Engineer Intern, Dot Spiders , Hyderabad ,India

Jul 2021 to Mar 2023

Project: Reactivities

Technologies: React.js, TypeScript, Vite, ASP.NET Core, C#, EF Core, SQLite, REST APIs, MediatR, AutoMapper, FluentValidation, Material UI, Axios, React Query, MobX, GitHub

- Developed full-stack web application features using **React.js, TypeScript, ASP.NET Core, C#, EF Core, SQL/SQLite, REST APIs, JavaScript, HTML, CSS, and Bootstrap.**
- Built a full-stack **activity management application** with React frontend and ASP.NET Core Web API backend for activity CRUD operations.
- Created responsive React UI features using **Material UI, React Router, Axios, TanStack React Query, MobX, and React Toastify.**
- Implemented layered architecture with **API, Application, Domain, Persistence, and Client** projects for maintainability and separation of concerns.
- Applied **MediatR/CQRS, AutoMapper, DTOs, FluentValidation, dependency injection, and custom exception middleware** to improve backend structure and error handling.
- Integrated **Entity Framework Core** for database access, migrations, data persistence, and startup data seeding.
- Debugged UI, backend, API, and database defects while collaborating with senior developers on code reviews, bug fixes, and feature enhancements.

EDUCATION

Eastern Kentucky University, Richmond

Master's degree in Computer Science • GPA: 3.88

Parul University

Bachelor's degree in Computer Science • GPA: 9.0